Arctic Wintertime Sea Ice Lead Detection From Sentinel-1 SAR Images

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***Abstract*— Leads are almost linear fractures within the ice pack, which are commonly observed in polar regions. In wintertime, leads promote energy flux from the underlying ocean to the atmosphere. Synthetic aperture radar (SAR) can monitor leads at a finer spatial resolution than other spaceborne datasets, regardless of solar illumination and atmospheric conditions. However, the SAR-based lead detection methods proposed to date are restricted to some specific areas, instead of the entire Arctic. In this article, we present a generalized deep learning-based approach for automatic sea ice lead detection (SILDET) in the Arctic wintertime using Sentinel-1 SAR images. The validation results show that SILDET has the capability of detecting open and frozen leads at different stages of development. Compared with the visual interpretation of Sentinel-1 images, the overall detection accuracy is 97.80% and the Kappa coefficient is 0.88. The lead map of a regional study obtained from SILDET was compared to that from a previous SAR-based lead detection method and a lead dataset based on Moderate Resolution Imaging Spectroradiometer (MODIS) data. The lead map was also validated using Sentinel-2 images. The result shows that SILDET can provide a more detailed distribution of leads and a better estimation of lead width and area. SILDET was applied to present the Arctic-wide lead distribution from January to April 2023 with a spatial resolution of 40 m. The Arctic-wide lead width distribution follows a power law with an average exponent of 1.65. The SILDET approach can be expected to provide long-term high-resolution lead distribution records.**

***Index Terms*— Arctic ocean, deep learning, sea ice lead, segmentation, Sentinel-1.**

I. Introduction

SEA ice leads are semi-linear fractures in the ice pack, which are usually formed during the divergence of ice cover. Lead width can vary from meters to kilometers. In wintertime, leads refreeze within a short time under the cold atmospheric temperature. Leads are composed of water and thin ice of a thickness of usually less than 30 cm (new and young ice, according to the World Meteorological Organization (WMO) [1]). Although open leads cover only 1%–2% of the total ice cover area in wintertime in the Arctic [2], they contribute more than 70% to the heat fluxes to the atmosphere [2]. Turbulent heat fluxes over open leads can be two orders of magnitude higher than those over the ice surface in winter [3]. The heat flux over a lead decreases as the frozen ice thickness increases, but is still significantly higher than that over the surrounding ice pack [4], [5]. Therefore, open and frozen leads play important roles in heat transfer over the Arctic Ocean, which further impacts the evolution of sea ice. Since Arctic sea ice has become thinner and more mobile in recent years [6], [7], [8], [9], an increasing role of leads in heat transfer over the Arctic Ocean can be expected. The areas of leads from midwinter to late spring could be a predictor for summer Arctic sea ice extent [10].

Satellite sensors can provide long time series and wide coverage for Arctic sea ice lead monitoring, with several products already available based on moderate-resolution thermal infrared images [11], [12], microwave images [13], [14], and radar altimetry [15]. However, the relatively coarse spatial resolution ($\geq$1 km) of the data from these datasets limits the detection of narrow leads and contributes to the overestimation of the width of leads [15], [16], [17], [18]. In addition, the frequent presence of cloud cover in the Arctic [19] restricts observation by thermal infrared images [11], [12].

Synthetic aperture radar (SAR) is capable of monitoring sea ice, independent of daylight and cloud coverage conditions, at a large scale and high spatial resolution. Murashkin et al. [20] and Murashkin and Spreen [21] used a random forest classifier to detect sea ice leads from Sentinel-1 dual-polarization images, with the algorithm based on polarization features and textural features derived from the gray-level co-occurrence matrix. Deep learning models, which can extract high-level feature information, are beneficial for various image analysis tasks [22], [23]. Advanced deep learning models have been widely used in detecting sea ice leads. For example, Murashkin and Frost [24] adjusted the UNet++ convolutional neural network architecture to classify sea ice types from Sentinel-1 images, including leads with low surface roughness and with wind-roughened surfaces; Zakhvatkina et al. [25] used texture characteristics as well as the difference and ratio between

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the co- and cross-polarizations (HH and HV) from Sentinel-1 images to detect leads; Murashkin [26] used U-Net [27] to detect leads from Sentinel-1 dual-polarization images, where, compared with the approach based on the random forest classifier [20], it obtained more robust results [26]; and Liang et al. [28] developed an algorithm based on an improved U-Net model [27]—the entropy-weighted network (EW-Net)—to detect open leads under low wind speed conditions from Sentinel-1 images. However, most of the existing SAR-based methods were tested on only a few scenes. To the best of our knowledge, there are no available Arctic-wide lead products based on the classification of SAR images, as a range of factors hinders the generalization and robustness of the SAR-based detection methods.

As sea ice leads cover only small parts of the ice cover area in the Arctic [2], the number of observations belonging to leads is significantly lower than that belonging to the other classes. With a deep learning approach, this class imbalance problem affects the convergence during the training phase and the generalization of the model in segmentation performance [29], [30]. In addition, the variations in thickness and surface roughness that are characteristics of frozen leads strongly affect the backscatter signature. Leads covered with calm open water have the lowest backscatter intensities. The backscatter coefficient (i.e., σ^o_{hh} in HH polarization) increases with thickness during the early growth of sea ice up to 30 cm thick [31], [32], [33]. Frost flowers, which are fragile and highly salty ice crystals, form on the surface of newly formed sea ice [34]. The presence of frost flowers creates high roughness and surface scattering, resulting in strong C-band HH backscatter [35]. The different backscatter variations of leads at the different stages of freezing cause difficulty in lead detection. Therefore, some of the previous lead detection studies focused on open leads or leads with low backscatter [20], [21], [26], [28]. Furthermore, the backscatter depends on the SAR incidence angle, which should be accounted for. Finally, although using dual-polarization images can improve the accuracy of lead detection, compared to using single-polarization images [20], the thermal noise in the HV polarization images restricts the generalization and robustness of the detection algorithms [36]. This is especially true when the lead is covered by a mixture of sea ice and water [37], [38].

In this article, we aim to enhance the generalizability and robustness of a deep learning-based method for sea ice lead detection (SILDET). We propose a robust and efficient approach for SILDET in the Arctic wintertime (January to April) from Sentinel-1 HH polarization images. Note that the openings between ice floes are also included under the term "lead." Three classes of surface cover are considered: 1) open leads; 2) frozen leads with thin ice (<0.30 m thick); and 3) background. The proposed method focuses on detecting the leads that exist in high ice concentration sea ice cover and with widths <30 km. To improve the performance of lead detection, the pyramid scene parsing network (PSPNet) [39] utilizing global scene category clues is applied. A new loss function is also proposed to alleviate the class imbalance problem. In addition, we utilize the narrow/elongated feature of leads to optimize the detection results. Finally, ancillary data are used to remove landfast ice and land, as well as the area outside the ice cover. In this article, we compare the lead maps from SILDET with those from visual interpretation and other existing lead detection methods. We then discuss the advantages and limitations of the proposed method. Furthermore, we describe how Arctic-wide lead distribution maps from January to April 2023 were generated utilizing SILDET. We then analyze the spatial distribution and width distribution of leads at a smaller scale than in previous studies.

II. Data

A. Sentinel-1 Images

Sentinel-1A/B is a constellation of two polar-orbiting satellites, developed and operated by the European Space Agency (ESA). The Sentinel-1A and Sentinel-1B satellites were launched in April 2014 and April 2016, respectively. Unfortunately, Sentinel-1B has not been available since December 23, 2021. Each satellite carries a C-band SAR system with dual-polarized channels. The sensor has four standard operational modes: stripmap (SM), interferometric wide swath (IW), extra wide swath (EW), and wave (WV). Sentinel-1 images are available free of charge from the Alaska Satellite Facility and Copernicus Data Space Ecosystem. In this study, we used the Level-1 ground range detected (GRD) product. All the scenes were acquired in EW mode from January to April over the Arctic sea ice. Only the HH polarization images were used in this study. The EW mode GRD product has an approximately 400-km swath width, with an incidence angle ranging from 18.9° to 47° and a spatial resolution of 93 × 87 m (pixel spacing of 40 × 40 m). The constellation has a combined repeat cycle of six days, with a greater frequency in the polar regions.

B. Ancillary Data

The following ancillary data were used in the SILDET: sea ice concentration (SIC) data, landfast ice maps, and a land mask. The SIC data were derived from the Advanced Microwave Scanning Radiometer-2 (AMSR2) using the Arctic Radiation and Turbulence Interaction Study (ARTIST) sea ice (ASI) algorithm, with a 6.25-km spatial resolution for the dataset [40]. The ASI SIC data were provided by the Institute of Environmental Physics (IUP) at the University of Bremen, Germany. In addition, the landfast ice maps were derived from the weekly ice charts produced by the U.S. National Ice Center (NIC). NIC ice charts provide information about ice types, including landfast ice distribution. The charts are based on the integration of SAR data, visible and infrared imagery, and meteorological data. In addition, a shapefile including land polygons was derived from the OpenStreetMap project.[1]

The 10-m wind parameters, as described in Section IV-A, were obtained from the ERA5 hourly data at single levels from 1979 to the present [41].

The ice drift speeds were derived using ice motion data from the National Snow and Ice Data Center (NSIDC), and are used for reference in Section IV-B.

[1]https://osmdata.openstreetmap.de/data/land-polygons.html

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C. Comparison Data

Murashkin [26] utilized the U-Net convolutional neural network to detect sea ice leads from both the HH and HV SAR channels of Sentinel-1 EW scenes. This algorithm produces binary maps with a lead classification at the same pixel spacing (40 m) as SILDET. Two types of leads are considered in this method. One type is the leads covered by water and thin ice, with a smooth surface and low backscatter in both HH and HV polarization images. The other type is the open leads with a rough surface affected by wind, leading to high backscatter in the HH images but low backscatter in the HV images. The lead fractions obtained from this method are close to the observed open-water fractions derived from airborne ice thickness observations [42] during the Multidisciplinary drifting Observatory for the Study of Arctic Climate (MOSAiC) expedition [43]. This implies that this method detects predominantly open water and leads covered by smooth thin ice.

Qu et al. [44] produced daily sea ice lead maps for April from 2001 to 2020 in the Beaufort Sea using temperature anomalies from Moderate Resolution Imaging Spectroradiometer (MODIS) thermal imagery. Two types are presented in the maps: lead (5% uncertainty) and a potential lead class (50% uncertainty). This dataset was compared with the other existing lead datasets and Landsat 8 images, and showed a better performance in identifying leads with open water and frozen thin ice. The MYD29 MODIS ice surface temperature (IST) daily L3 product with a spatial resolution of 1 km is used for reference in the lead map comparison in Section IV-B.

A lead map was generated from Sentinel-2 images based on a threshold of reflectance. The Sentinel-2 mission comprises a constellation of two polar-orbiting satellites in the same sun-synchronous orbit, with a ten-day repeat cycle for each satellite. The payload of the Sentinel-2 satellites is a single MultiSpectral Instrument (MSI) composed of 13 bands, including blue, green, and red channel images (bands 2–4). The true-color (red: band 4, green: band 3, blue: band 2) composite images with a spatial resolution of 10 m were used in this study. We downloaded the Sentinel-2 Level-2A surface reflectance images from the Copernicus Open Access Hub.

In this article, for evaluation purposes, we compare the lead map from SILDET for April 20, 2019 in the Beaufort Sea with three other maps from the datasets mentioned above.

III. Methodology

In this section, we present a detailed description of the proposed method for SILDET from Sentinel-1 HH polarization SAR imagery. The description covers the preprocessing of SAR images, the generation of the training dataset, the settings for the training models, the framework of SILDET, and the evaluation of the detection results. A flowchart for the SAR image preprocessing and SILDET is illustrated in Fig. 1. SILDET consists of four modules: 1) a segmentation module; 2) a balance module; 3) an optimization module; and 4) a mask module.

A. Sentinel-1 Image Preprocessing

The Sentinel-1 image preprocessing included border noise removal, thermal noise removal, radiometric calibration with

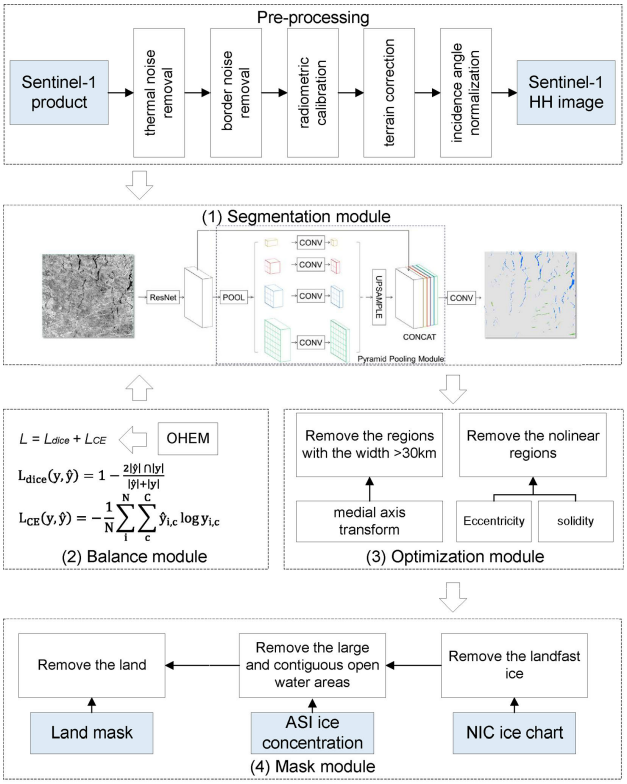


Fig. 1. Flowchart of the preprocessing and the proposed lead detection method (SILDET).

conversion to decibels, terrain correction, and incidence angle normalization. All the preprocessing was performed in the sentinel application platform (SNAP), except for incidence angle normalization. The correction factors that account for the incidence angle are different for sea ice and water [45]. We analyzed the dependence of the incidence angle for sea ice and water using 480 sample points for each class from a set of 24 Sentinel-1 scenes acquired in the Arctic from January to April over the period of 2016–2022 (Table VIII in Appendix A). Considering the effect of wind on the backscatter in HH images, we visually selected samples of wind-free water surfaces in the sea ice leads. The variation of backscatter for ice and water with respect to incidence angle is shown in Fig. 2. The incidence angle slopes of the linear fits for sea ice and water are -0.26 and -0.11 ($p < 0.05$), respectively. These values are comparable to those reported in previous studies [36], [46]. We used a threshold to determine which incidence angle slope should be used for a pixel. A threshold was set using -20 dB for σ^o_{hh} to effectively separate ice and water (ice, 86% accuracy; water, 85% accuracy). The resulting equation for the incidence angle normalization is

$$\begin{aligned}\sigma^o_{\mathrm{hh-corr}} &= \sigma^o_{\mathrm{hh}} + k_1 \cdot (\theta - \min(\theta)) \quad \text{if } \sigma^o_{\mathrm{hh}} \geq -20 \\ \sigma^o_{\mathrm{hh-corr}} &= \sigma^o_{\mathrm{hh}} + k_2 \cdot (\theta - \min(\theta)) \quad \text{if } \sigma^o_{\mathrm{hh}} < -20 \end{aligned} \tag{1}$$

where σ^o_{hh} is the backscatter coefficient of HH; $\sigma^o_{\mathrm{hh-corr}}$ is the backscatter coefficient of HH after correction, given in decibels; and θ is the incidence angle in degrees. $k_1 = 0.26$ and $k_2 = 0.11$.

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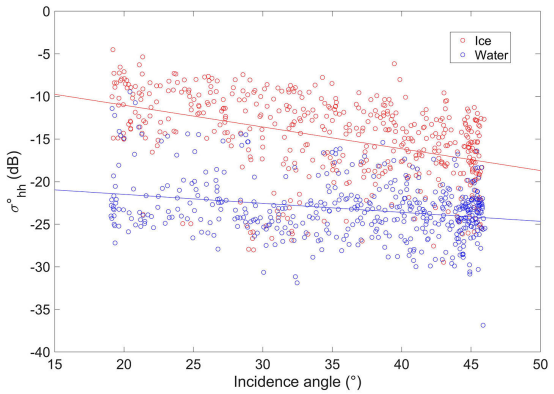


Fig. 2. Variation of the backscatter (σ^o_{hh}) from sea ice and open water with incidence angle across the swath of the SAR imagery.

TABLE I
DETAILS OF THE TRAINING DATASET

Sample category	Open lead	Frozen lead	Background
Total pixels	9.758×10^6	4.075×10^7	7.935×10^8

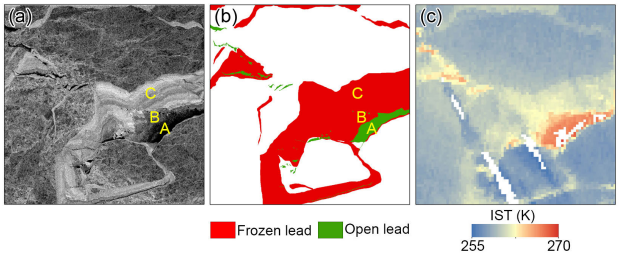


Fig. 3. Subimage of 2048 × 2048 pixels in the training dataset. (a) Sentinel-1 HH polarization image was acquired at 15:11 on April 16, 2017. (b) Corresponding categories are obtained by visual interpretation. (c) MODIS IST image acquired at 21:54 on April 16, 2017. Note that the white areas in (c) indicate cloud coverage.

B. Generation of the Training Dataset and Settings for the Training Models

The sea ice lead training dataset consisted of subimages of Sentinel-1 HH polarization and the corresponding categories. Each subimage was 2048 × 2048 pixels in size. A total of 211 subimages from 23 scenes were used to train the segmentation networks (Table IX in Appendix A), with about 5–15 subimages derived from each scene. Three surface categories were used in the training: 1) open lead; 2) frozen lead; and 3) background (Table I). Visual interpretation was performed based on the Sentinel-1 images and MODIS IST images. The following criteria were used.

1) *Open Leads:* These are openings within the ice cover or between ice floes with the lowest backscatter intensities (see the region with label A in Fig. 3). Although the backscatter intensity of open water depends on the surface wind, the wind effect on leads within sea ice cover is less than that over the open ocean, due to the wind shadowing by ice floes [47]. Therefore, we do not consider the wind effect on the identified leads in this article.
2) *Frozen Leads:* The leads with new ice (<0.10 m thick) and young ice (0.10–0.30 m thick), according to WMO nomenclature [48], were labeled as "frozen lead." A frozen lead surface usually has increasing backscatter as the new ice thickness increases [49], [50] (see the region with label B in Fig. 3). The backscatter of the leads at the different stages of development is different. There is often a "stream line" texture (see the region with label C in Fig. 3) representing different ice growth stages in the frozen leads. In addition, all the frozen leads are characterized by a linear shape and higher IST than the surrounding ice pack. MODIS IST data were used to confirm the manual interpretation of leads in the SAR images, as shown in Fig. 3(c).
3) *Background:* This category included all objects that did not lead. Ice floes in leads were also classified as "background." The training dataset consisted of scenes with open and frozen leads at different development stages.

Considering the computational cost, the samples from the training dataset were cropped from 2048 × 2048 pixel patches to 512 × 512 pixel patches with an overlap of 50%. To improve the generalization capability of the model and to prevent overfitting, we used random horizontal and vertical flipping as data augmentation for these samples with 512 × 512 pixels. Finally, a total of 15 192 samples of 512 × 512 pixels were used in training the models.

We employed the PyTorch framework to implement the networks, which were trained on an NVIDIA GeForce RTX 3090 Ti GPU. The training was conducted for 100 epochs with a batch size of 16. The Adam optimization algorithm [51] was selected as the optimizer. The initial learning rate was set as 0.0001 and was decreased by a ratio of 0.91 after each epoch.

C. SILDET Approach

The components of SILDET (see Fig. 1) are explained in the following.

1) Segmentation Module: The segmentation module uses PSPNet [39] to obtain the initial segmentation result from the Sentinel-1 input images, which are 512 × 512 HH polarization backscatter images. A ResNet [52] model is used to extract the feature map. The top feature map size is 1/8 of the input image size. We used the ResNet-101 model as the backbone in this study. After this, a pyramid pooling module gathers contextual information from the top feature maps. This module enables PSPNet to collect levels of information and increases the receptive field for more accurate segmentation results. The pyramid pooling module consists of four global adaptive average pooling layers to obtain feature maps at various scales (1 × 1, 2 × 2, 3 × 3, and 6 × 6). The dimension of these feature maps is then reduced to $1/N$ using a 1 × 1 convolutional layer, where N represents the level number, which is set to 4. In addition, the feature maps are expanded through bilinear interpolation. These feature maps and the input feature map from ResNet are finally flattened and concatenated for classification. A convolutional layer is used to generate the final prediction map.

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2) Balance Module: The training samples in the training dataset are extremely imbalanced among the different classes listed in Table I, which could have degraded the performance of the networks. To deal with this problem, data augmentation methods are usually employed, such as geometrical transformations of the images (e.g., horizontal flipping and multiscale strategy) or using a random occlusion mask on the images or feature maps. On the other hand, some loss functions have been shown to be able to tackle class imbalance. We use dice loss as well as cross-entropy (CE) loss with online hard example mining (OHEM) to train the model in SILDET. CE loss is the entropy that calculates the prediction probability and the real situation based on each pixel in the dataset. Dice loss is different from CE loss in that the former is based on the dice overlap coefficient between the predicted results and real situations [53], and can thus alleviate class imbalance. OHEM is a training strategy that focuses on hard samples during the training process and applies higher weights to them [54]. This strategy can enable the model to concentrate more on hard samples, which can be applied to further optimize the CE loss function and make the training more effective and efficient. The hard sample pixels are determined by a probability threshold θ for the corresponding target class, and thus the pixels below the threshold are preserved in the training. The value of θ was set to 0.9 in this study.

The overall loss function for the segmentation module is defined as

$$L = L_{\text{dice}} + L_{\text{CE}} \tag{2}$$

$$L_{\text{dice}}(y, \hat{y}) = 1 - \frac{2|\hat{y}| \bigcap |y|}{|\hat{y}| + |y|} \tag{3}$$

$$L_{\text{CE}}(y, \hat{y}) = -\sum_{i} y \log \hat{y} \tag{4}$$

where y is the label and $\hat{y}$ is the corresponding prediction result. OHEM is applied in the calculation of the CE loss during training.

3) Optimization Module: The optimization module quantifies features to identify the surviving regions as leads in the initial segmentation result. First, the objects with a width $>$30 km are removed. This value was referred from [44]. The medial axis distance is used in this step. The medial axis of an object is the set of all the points having more than one closest point on the object's boundary [55]. Medial axis transform is used to compute the width of the objects. The ice floes misclassified as frozen leads are then removed. Sea ice leads have linear-like structures. In this step, the eccentricity and solidity are used. Eccentricity is the ratio of the distance between the foci of the ellipse and its major axis length. The value is between 0 and 1. An ellipse with an eccentricity is 0 is a circle, while an ellipse with an eccentricity of 1 is a line segment. However, a circular curve or circle would have similar eccentricity, so solidity is used as an additional feature to distinguish them. Solidity represents the ratio of pixels in the region to pixels of the convex hull image. According to the statistics based on the training dataset (Fig. 13 in Appendix B), the objects with eccentricity $<$0.95 and solidity $>$0.55 are not considered to be frozen leads.

TABLE II
LIST OF THE SENTINEL-1 SCENES SELECTED FOR EVALUATION OF THE SILDET RESULTS

No.	Acquisition date	Acquisition time	Coverage	Satellite
1	Mar. 8 2021	15:10:12	Beaufort Sea	Sentinel-1B
2	Apr. 20 2019	15:42:44	Beaufort Sea	Sentinel-1B
3	Apr. 15 2019	15:35:35	Beaufort Sea	Sentinel-1B
4	Jan. 4 2022	17:23:43	Chukchi Sea	Sentinel-1A
5	Feb. 22 2018	20:08:01	East Siberian Sea	Sentinel-1A
6	Feb. 25 2021	23:16:26	Laptev Sea	Sentinel-1A
7	Mar. 3 2022	2:51:27	Kara Sea	Sentinel-1A
8	Jan. 6 2021	7:53:57	Greenland Sea	Sentinel-1B

4) Mask Module: In this module, land, landfast ice, and the area outside the ice cover in the SAR images are removed. The land is removed using the land shapefile. In the coastal zones, landfast ice has a smooth surface and is often misclassified as open lead due to its similarity in low backscatter [56]. Landfast ice is masked using maps from the NIC ice charts. Daily ice cover maps from AIS SIC data are used to mask out the large and contiguous open-water areas. The daily ice cover is identified through three steps. In the first step, the regions with SIC $>$87% are generated to initiate the ice cover mask. This approach was based on the study by Guo et al. [46]. In order to eliminate the effects of ice motion during the time gap between SIC data and SAR data, the edge of the mask is reduced by 12.5 km in the second step. In the third step, leads with low ice concentration in the pack ice are filled using morphological operations. This process employs the hole-filling function, which identifies and fills enclosed background regions, ensuring a continuous ice cover. Lead detection results outside the ice cover maps are removed.

D. Evaluation of SILDET

To evaluate the performance and robustness of SILDET, we compared the SILDET results with Sentinel-1 visual interpretation. For comparison, eight subimages were obtained from different locations (Table II), which are shown in Fig. 4. Scenes (red boxes) from the Beaufort Sea, Chukchi Sea, East Siberian Sea, Laptev Sea, Kara Sea, and Greenland Sea were selected. In each month from January to April, two scenes were acquired. Finally, to test the robustness of the method, we ensured that the selected scenes covered different stages of lead freezing and sea ice conditions.

In this article, to quantitatively characterize the comparison between the SILDET results and the visual interpretation of Sentinel-1, we use four parameters: precision, recall, Kappa

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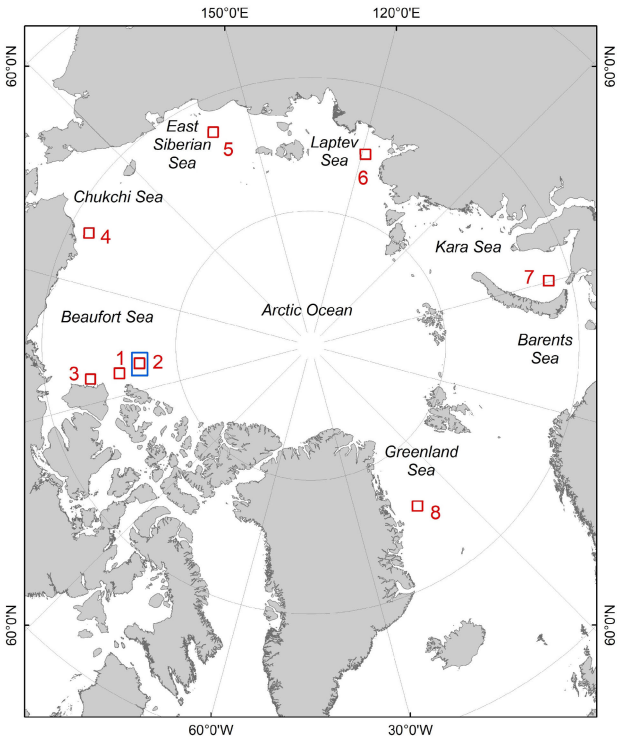


Fig. 4. Locations of the Sentinel-1 SAR subimages used for evaluation (red box) and the mosaic of Sentinel-2 images from which a lead map was generated for the purpose of comparison (blue box). The details of these scenes are presented in Table II.

coefficient, and the overall accuracy (OA), based on the confusion matrix [57], [58]. The confusion matrix is formulated from four indices: true positives (TPs), true negatives (TNs), false positives (FPs), and false negatives (FNs). Precision is the number of correct results (TP) relative to the total number of results. Recall is the number of correct results relative to the number of visual interpretation results. OA denotes the proportion of correctly classified pixels in all the pixels. The Kappa coefficient [59] is a useful evaluation metric when dealing with imbalanced data

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{5}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{6}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{7}$$

$$\text{Kappa} = \frac{\text{OA} - \text{Pe}}{1 - \text{Pe}} \tag{8}$$

where

$$\text{Pe} = \frac{(\text{TP} + \text{FP}) \times (\text{TP} + \text{FN}) \times (\text{FN} + \text{TN}) \times (\text{FP} + \text{TN})}{(\text{TP} + \text{FN} + \text{FP} + \text{TN})^2}. \tag{9}$$

To further assess the reliability of the SILDET results obtained by SILDET, a lead map from Sentinel-2 scenes was generated for comparison. The blue box in Fig. 4 shows the location of the Sentinel-2 mosaic in the Beaufort Sea, which is characterized by high lead frequency [44].

E. Lead Width and Length Statistics

We analyzed the lead width distribution from SILDET and the other lead detection methods. Previous studies using submarine measurements and remote sensing data have shown that the lead width distribution follows a power-law distribution [60], [61], [62], [63]:

$$N(x) = ax^{-b} \tag{10}$$

where $N(x)$ is the number of leads with width x and a and b are coefficients that can be determined from empirical data. A higher exponent (b) implies a higher fraction of narrow leads.

The method presented in Qu et al. [18] was used to calculate the lead width and length from the lead maps. The width for every pixel was determined by an orthogonal system (vertical and horizontal). However, assuming a parallel lead boundary, this method tends to overestimate lead width by up to 41.4% [18]. The overall length L_i for lead width X_i can be calculated as follows:

$$L_i = \frac{a_0^2 N_i}{X_i} = \frac{a_0 N_i}{i} \tag{11}$$

where a_0 is the pixel resolution, and N_i is the number of pixels for width $X_i = a_0 \times i$, $(i = 1, 2, 3, \ldots)$.

Note that the two types—"open lead" and "frozen lead"—in the lead detection results are combined as one type in the lead width and length statistics.

IV. Results

A. Comparison With Visual Interpretation of Sentinel-1

Fig. 5 shows subimages from the four Sentinel-1 HH polarization scenes (Nos. 2, 3, 5, and 7), the locations of which are shown in Fig. 4, along with the corresponding lead identification obtained using visual interpretation, as well as the SILDET results. The results for the other scenes (Nos. 1, 4, 6, and 8) are shown in Fig. 12. The scenes cover different lead conditions with different stages of development. The numerical indicators that characterize the comparison between SILDET and the visual interpretation results are presented in Table III. The OA and Kappa coefficients for the SILDET detection results are 97.80% and 0.88, respectively, which means a good match between SILDET and the visual interpretation results. In addition, the frozen lead class has a precision of 95.66% and a recall of 79.05%, which indicates that the detected frozen leads are mostly classified correctly by SILDET, but some frozen leads in the visual interpretation images are ignored. For the open leads, the precision and recall values are 83.17% and 90.58%, respectively. The low recall of frozen leads and the precision of the open leads can be mainly attributed to the misclassification of some frozen leads as open leads, because of their low backscatter. In general, SILDET can correctly detect most of the visible open and frozen leads from the SAR images. However, it does still generate some incorrect identifications due to backscatter overlapping.

Scene No. 7 in Fig. 5(a) shows the first-year ice (FYI) and open leads in the Sentinel-1 imagery. Most of the visible open leads are detected using SILDET. The evaluation parameters

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TABLE III

PRECISION, RECALL, KAPPA COEFFICIENT, AND OA FOR THE SILDET-BASED LEAD DETECTION RESULTS IN COMPARISON TO VISUAL INTERPRETATION OF SENTINEL-1 IMAGES. THE NUMBERING ORDER OF THE SCENES IS CONSISTENT WITH TABLE II

Scene No.	Precision (%)		Recall (%)		Kappa coefficient	Overall accuracy (%)
	Open lead	Frozen lead	Open lead	Frozen lead		
1	76.98	-	90.04	-	0.82	99.24
2	91.16	97.77	88.52	87.97	0.91	96.73
3	67.22	94.50	92.12	81.36	0.85	95.64
4	84.08	86.67	89.07	2.22	0.82	98.04
5	85.26	86.57	83.44	71.37	0.80	98.33
6	81.30	0.00	95.57	0.00	0.86	97.43
7	86.87	-	81.07	-	0.83	98.44
8	87.39	-	93.94	-	0.90	98.51
Overall	**83.17**	**95.66**	**90.58**	**79.05**	**0.88**	**97.80**

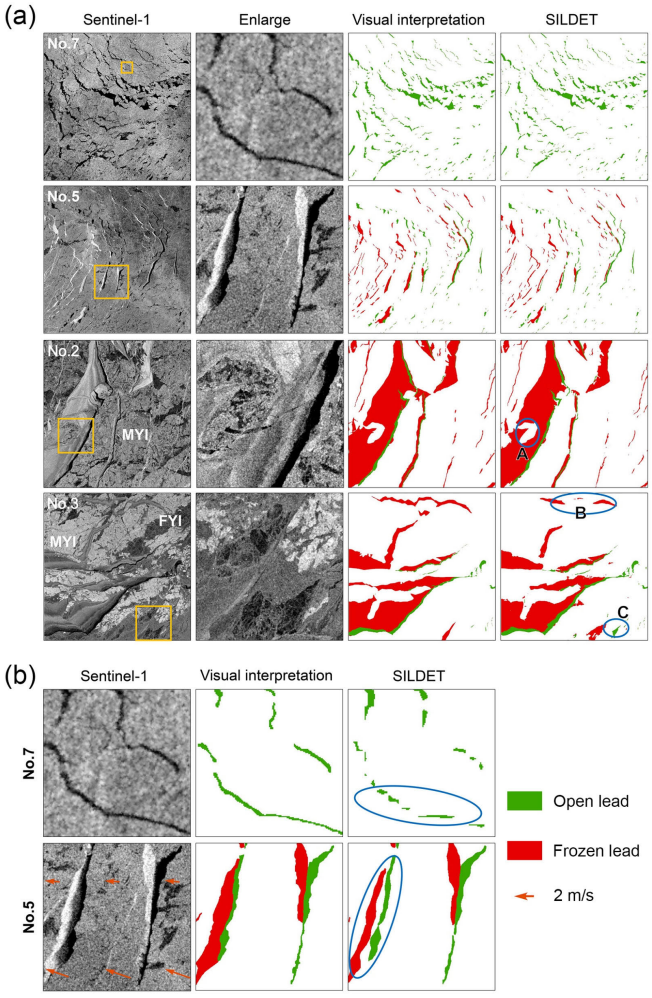


Fig. 5. (a) Four Sentinel-1 σ^o_{hh} images of scenes (first column). Enlargements of the areas in the yellow boxes are shown in the second column. Lead identification based on visual interpretation of SAR images and the SILDET method is shown in the third and fourth columns, respectively. (b) Enlargement of the lead identification using visual interpretation and SILDET for the enlarged areas in scenes 7 and 5 in (a). The orange arrows represent wind data. The length of the arrows represents the value of the wind speed.

for this scene are high (Table III). However, some narrow leads with width <5 pixels are incompletely detected, such as the area marked by the blue ellipse in Fig. 5(b). The open leads present as discontinuous punctiform or linear objects in the SILDET detection results.

For Scene No. 5, many of the leads contain frozen sections. Most of the open and frozen sections are identified by SILDET. Fig. 5(b) shows an enlargement of the area in the yellow box in Scene 2. The average 10-m wind speed from ERA5 at 20:00 GMT was 4.31 m/s. In the area marked by the blue ellipse in Fig. 5(b), there is likely thin ice forming in the leads, which is being advected downwind. Some frozen parts of the leads with similar backscatter and texture to the surrounding FYI are ignored by SILDET. This results in a decrease in the recall for the frozen leads to 71.37%.

Scenes 2 and 3 (Table II) feature multiyear ice (MYI), FYI, open leads, and frozen leads. In Scene 3, there are ice floes within the frozen leads, as shown in the area with label A. Although the ice floes have a backscatter similar to that of the surrounding frozen leads, their texture is different. Both SILDET and visual interpretation can distinguish the ice floes from the background of frozen leads well (areas with white within the blue ellipse with label A). This can be attributed to the global context utilized by PSPNet in SILDET. In Scene No. 4, some frozen lead pixels are not detected (the blue ellipse with label B), due to their small backscatter difference with the surrounding ice. In addition, smooth FYI is misclassified as open lead in the area marked by the blue ellipses with label C. This causes the precision to take on a relatively low value of 67.22% for the open leads, which is the lowest value in Table III.

B. Comparison With the Lead Maps From Other Methods

To further assess the reliability of the SILDET results obtained by SILDET, we compared the lead results overlaid on a mosaicked true-color image assembled from four Sentinel-2 scenes (the location of each Sentinel-2 scene is shown in Fig. 13 in Appendix D). The four Sentinel-2 scenes were acquired at 21:22 on April 20, 2019, 22:31 on April 21, 2019, and 21:40 on April 21, 2019 in the Beaufort Sea. The locations of the frames are shown in Fig. 4 (blue box). Fig. 6(d)–(g) shows examples of the lead detection maps from the four methods: d) SILDET based on Sentinel-1 images; e) Murashkin [26] method based on Sentinel-1 images; f) the

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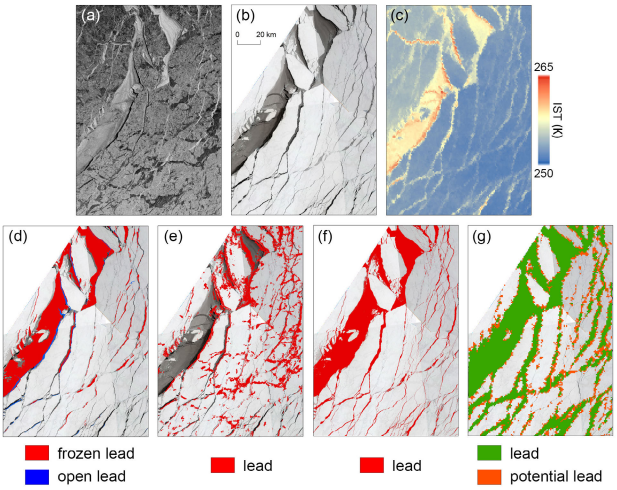


Fig. 6. Images and leads maps for April 20/21, 2019 in the Beaufort Sea. (a) Sentinel-1 σ^o_{hh} image. (b) Sentinel-2 true-color composite image (red: band 4, green: band 3, and blue: band 2). (c) MODIS IST Aqua daily L3 image. (d) Lead map from the SILDET algorithm based on Sentinel-1. (e) Lead map from the Murashkin method based on Sentinel-1. (f) Lead map from Sentinel-2. (g) Lead map from the Qu et al. dataset.

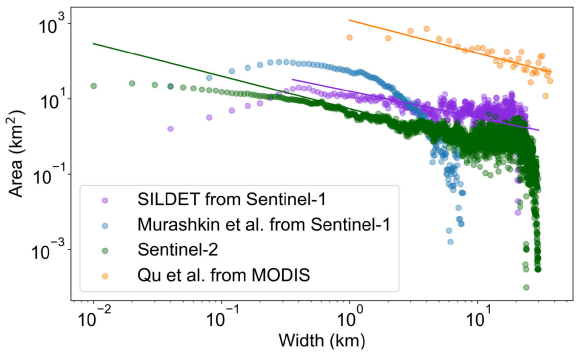


Fig. 7. Log–log plot of the lead width distribution from the lead maps (the frame of Fig. 6) for April 20/21, 2019 in the Beaufort Sea. Straight lines are the fit power-law curves. Note that power-law fitting from Sentinel-1 applies only for a width of 360 m and more.

method based on a threshold for the reflectance in Sentinel-2 images; and g) the Qu, et al. [44] method based on MODIS images. These maps are referred to as Fig. 6(d)–(g) in the subsequent parts of this section. All the lead maps are overlaid on the Sentinel-2 images.

The Sentinel-1 scene from 15:42 on April 20, 2019 is shown in Fig. 6(a). The time difference between the acquisition of the Sentinel-1 and Sentinel-2 scenes ranges from 6 h to one day. The average ice drift speed on April 20, 2019 within the frame of Fig. 6 was 1.71 km/day. In this case, the open and frozen leads are clearly visible in the pack ice region. The open leads are represented by low backscatter in the HH polarization images. The frozen leads show a wide range of σ^o_{hh}, due to differences in the stages of freezing and the surface characteristics. A few ice floes can be observed in some of the frozen leads. The Sentinel-2 mosaicked image is shown in Fig. 6(b), where the reflectance from leads is lower than that from the surrounding ice. The MODIS IST Aqua daily L3 image from April 20, 2019 is shown in Fig. 6(c). We generated the lead map [see Fig. 6(f)] from the Sentinel-2 mosaicked image using the reflectance from band 2 (490 nm) with a threshold of $<$70%. The threshold was determined after analyzing the reflectance distribution of the image (see Fig. 14(a) in Appendix D). Some leads with low reflectance are not identified. The open leads and frozen leads in this map are not distinguished. All the lead maps in Fig. 6 are clipped to the same extent as the frame of the Sentinel-2 scene in Fig. 6(b).

Compared with the Sentinel-2 images, SILDET captures the majority of the leads and shows the detailed spatial distribution of open leads, frozen leads, as well as ice floes in leads. Although the open leads shown in the SAR image could have frozen during the Sentinel-2 overpass, the open leads in our detection results are consistent with the areas of low reflectance in the Sentinel-2 images. Meanwhile, Fig. 6(d) is also basically consistent with the area that has a relatively high IST in the MODIS image. However, we found that SILDET misses some small leads in the Sentinel-2 images. The reason for this could be the limitations of SILDET, the spatial resolution differences, or ice motion due to the time difference between the data acquisition. Fig. 6(e) shows more small leads but fewer frozen leads than Fig. 6(f). Meanwhile, some FYI with low backscatter is misclassified as leads. The lead detection results of Fig. 6(g) identify most of the leads, compared to the Sentinel-2 retrievals. However, it appears that the edges of the leads are beyond the bounds of the leads in the Sentinel-2 images. This is probably because this dataset is based on MODIS thermal infrared images from many orbits within the same day.

We estimated and compared the lead width distributions of these lead maps. Note that we only include the "lead" type in Fig. 6(g) in this section. Fig. 7 shows the width distribution of the leads and their power-law fittings (the frame of Fig. 6). The lead width distribution from Fig. 6(d) is close to that from Fig. 6(f). However, SILDET has limitations in detecting narrow leads. For the narrow leads, the maximum area is associated with a lead width of around 360 m (9 pixels). A possible source of error in the distribution could be the receptive field of PSPNet with a theoretical size of 8×8. A large receptive field is not suitable for small object detection (smaller than 8×8). Meanwhile, the mixed pixels at the edges of the leads also affect the identification of narrow leads. We removed the data with a width $<$360 m in Fig. 6(d) to fit a power law. The method proposed by Murashkin [26] was designed to detect open leads or frozen leads with smooth surfaces. As a result, this method has limitations in detecting frozen leads, and Fig. 6(e) therefore shows a different lead width distribution, compared with the other methods. Compared with Fig. 6(f), the result in Fig. 6(e) shows more leads with small widths and fewer leads with large widths. For the leads of all widths in Fig. 6(g), the areas of the leads are overestimated. This difference may be due to the mixed pixels and the coarse spatial resolution of the MODIS images.

We compared the power-law fittings of the lead width "x" of the three lead maps Fig. 6(d)–(g). They are presented as follows. Note that the result from Fig. 6(e) does not fit the power-law function.

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Fig. 6(d): $N(x) = 15.98\ x^{-0.70}$ (0.36 km $\leq x \leq$ 25.28 km).
Fig. 6(f): $N(x) = 5.48\ x^{-0.86}$ (0.01 km $\leq x \leq$ 29.83 km).
Fig. 6(g): $N(x) = 1219.20\ x^{-0.87}$ (1 km $\leq x \leq$ 37 km).

The SILDET result from Sentinel-1 has the lowest power-law exponent, which is likely due to the limitation of SILDET in detecting narrow leads. The lead map from Qu et al. [44] has the lowest spatial resolution among the three sensors but the highest power-law exponent for the data. It is possible that the lead retrieval method from MODIS images is sensitive to narrow leads with a width smaller than the pixel width (<1 km). The power-law exponent of Sentinel-2 is close to that for MODIS. The methods for detecting leads from the MODIS and Sentinel-2 images are based on thresholds, and show better results for narrow leads, whereas SILDET is based on a segmentation network. The limitations of the segmentation network approach are discussed in Section V-B.

Furthermore, we list the length and area of the lead categories with different width ranges for these three methods in Table IV. We calculated the lengths and areas for the five categories of leads as (see Table IV): width $\leq$ 100 m, 100 m < width $\leq$ 500 m, 500 m < width $\leq$ 1 km, 1 km < width $\leq$ 5 km, and width > 5 km for the frame shown in Fig. 6. These are denoted as A, B, C, D, and E, respectively.

The total length of the leads from Fig. 6(d) is 1254.38 km. The Category B leads contribute most to the total length, with about 36%. The total length of the leads from Fig. 6(f) is 9209.64 km, with the Category A leads constituting about 70% of the total number of leads. The total length of the leads from Fig. 6(g) is close to that of Fig. 6(d), with 1415.46 km. Fig. 6(d) and (g) clearly underestimate the total length of the leads, compared with Fig. 6(f). The main reason for this is that the lengths of the Category A and B leads are underestimated in SILDET and ignored in the method of Qu et al. [44]. Although Sentinel-1 has a nearly 100-m resolution, the measured lead Category A accounts for only 6.44% of the total length. MODIS has a low resolution and cannot provide information for these narrow leads. Fig. 6(g) is sensitive to the small leads with a width $\leq$1 km, which accounts for 30.45% of the total length. It is possible that the method of detecting leads from MODIS images can detect subpixel leads because it relies on thresholds for the surface temperature anomalies, which can detect a lead covered by only part of a pixel.

As for the lead area from the three lead maps, Fig. 6(d) and (f) have close values of the total lead area, with 2922.65 and 3792.50 km^2, respectively. The total area of leads is overestimated by 68% in Fig. 6(g), compared to Fig. 6(f). For the three lead maps, the highest contributions are from the Category E leads, with more than 50%. For the Categories A and B leads, Fig. 6(d) shows an underestimation of their contribution to the total lead area, compared with Fig. 6(f). For the lead categories with widths $\geq$1 km, the contributions to the total lead area of the different lead categories are similar among the three lead maps.

C. Application of SILDET

To further explore the applicability of SILDET, we generated Arctic-wide lead frequency (including open and frozen leads) maps based on Sentinel-1 data from January to April 2023 with a spatial resolution of 40 m (Fig. 8). Lead frequency is defined as the number of scenes during a specified period (a month for this figure) in which a pixel is found to be covered by lead, relative to the number of available Sentinel-1 scenes. A total of 1108, 1045, 1169, and 1139 Sentinel-1 scenes were used to generate the maps for each month from January to April 2023, respectively. The image of the Beaufort Sea is enlarged in Fig. 9. The spatial distribution of the number of Sentinel-1 scenes from January to April 2023 is shown in Fig. 15 in Appendix E. Due to the retirement of the Sentinel-1B satellite in December 2021, there are fewer scenes than before in the Arctic. However, most of the regions in the area covered by the Sentinel-1 images have more than ten coverages per month from January to April 2023.

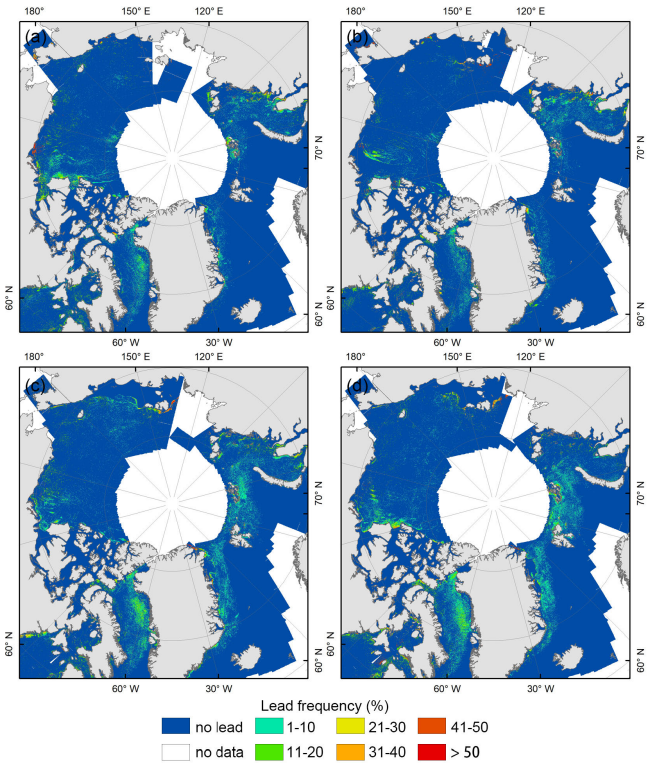


Fig. 8. Spatial distribution of the 40-m sea ice lead frequency from January to April 2023. (a)–(d) Arctic. The white areas indicate no Sentinel-1 data.

Fig. 8 shows that the lead frequency is less than 20% in most Arctic regions. This should be considered in the context of rapid ice motion during the time between observations and the short lead lifetime. Regions with a relatively high lead frequency include the Beaufort Sea, Baffin Bay, Fram Strait, Kara Sea, and the Barents Sea. The relatively high lead frequency in the Beaufort Sea is found in regions closer to land. In addition, compared with January and February, the lead frequency increases in March and April, as found in Baffin Bay, Fram Strait, and the Barents Sea. These regions are close to the marginal ice zone, where the pack ice becomes less dense in March and April.

The lead width distributions from January to April 2023 in the Arctic and Beaufort Sea are presented in Fig. 10. The number of pixels for each lead width is fit with a power-law

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TABLE IV

LENGTH AND AREA OF THE RETRIEVED LEADS FROM THE LEAD MAPS BASED ON SENTINEL-1, SENTINEL-2, AND MODIS (THE FRAME OF FIG. 6) ON APRIL 20/21, 2019 IN THE BEAUFORT SEA, CATEGORIZED BY WIDTH (a: ≤100 m, b: 100–500 m, c: 500 m–1 km, d: 1–5 km, AND e: >5 km) NOTE: THE DATA WITH LEAD WIDTH >30 km FROM SILDET ARE NOT CONSIDERED IN THE STATISTICS

Satellite and method	Lead category	Lead length		Lead area	
		(km)	Contribution (%)	(km^2)	Contribution (%)
Sentinel-1 (SILDET)	A	80.78	6.44	4.85	0.17
	B	448.71	35.77	136.37	4.67
	C	238.42	19.01	171.04	5.85
	D	309.42	24.67	667.22	22.83
	E	177.05	14.11	1943.18	66.49
	Total	**1254.38**		**2922.65**	
Sentinel-2 (Threshold method)	A	6408.05	69.58	200.34	5.28
	B	1809.64	19.65	425.27	11.21
	C	430.10	4.67	300.96	7.94
	D	400.40	4.35	865.83	22.83
	E	164.77	1.79	2000.09	52.74
	Total	**9209.64**		**3792.50**	
MODIS (Qu *et al.* method)	A	-	-	-	-
	B	-	-	-	-
	C	431.00	30.45	431.00	6.73
	D	680.58	48.08	2177.00	34.02
	E	303.88	21.47	3792.00	59.25
	Total	**1415.46**		**6400.00**	

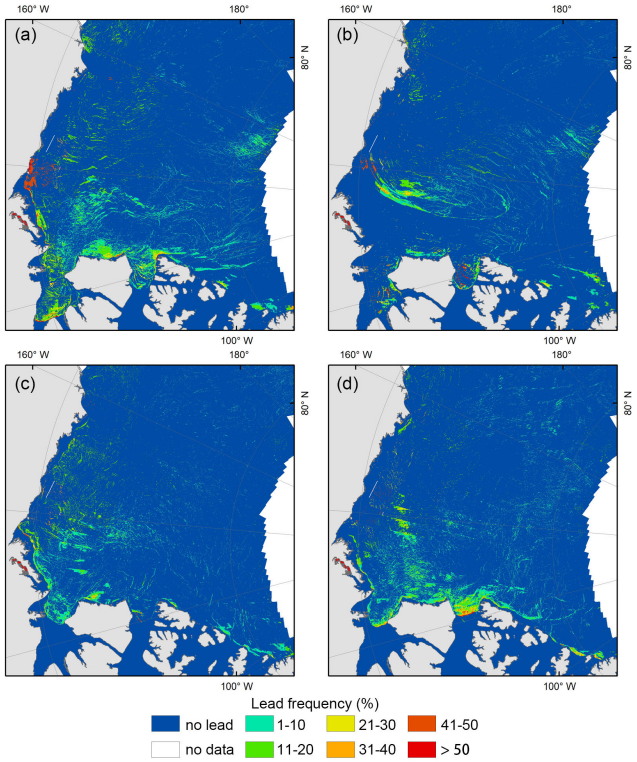


Fig. 9. Spatial distribution of the 40-m sea ice lead frequency from January to April 2023. (a)–(d) Beaufort Sea. The white areas indicate no Sentinel-1 data.

distribution. Considering the limitation of SILDET in detecting narrow leads, the points with widths <360 m are not included in the power-law fitting (see Section IV-B). The exponents of the power-law fitting for the lead width distribution are

TABLE V

EXPONENT OF THE POWER-LAW FITTING [(9) FOR $0.36\ \text{km} \leq x \leq 30\ \text{km}$] IN FIGS. 8 AND 9

Region	Month	Exponent	R^2
Arctic Ocean	January	1.57	0.97
	February	1.64	0.94
	March	1.69	0.97
	April	1.71	0.95
	mean	1.65	-
Beaufort Sea	January	1.61	0.92
	February	1.96	0.73
	March	1.89	0.87
	April	1.49	0.92
	mean	1.74	-

listed in Table V. For the Arctic Ocean, the differences in the lead width distribution between January, February, March, and April 2023 are small, while the exponent increases toward early spring with an average of 1.65. This could be related to the changes in internal ice strength. A tendency of the exponent to increase throughout the winter season during the MOSAiC expedition was also reported [64]. Regarding the Beaufort Sea, the lead width distribution across the four months mainly differs in the leads with large widths. In addition, the exponent varies from 1.49 to 1.96.

We further analyzed the length and area of the leads in January and April. The leads were divided into five categories, which were the same as those in Section IV-B. The length and area for these categories are presented in Table VI.

For the Arctic, compared with January, the detected lead length and area in April are higher by 42% and 44%,

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TABLE VI

LENGTH AND AREA OF THE RETRIEVED LEADS BASED ON SILDET FOR JANUARY AND APRIL 2023 IN THE ARCTIC AND BEAUFORT SEA, CATEGORIZED BY WIDTH (a: ≤100 m, b: 100–500 m, c: 500 m–1 km, d: 1–5 km, AND e: >5 km). NOTE: THE DATA WITH LEAD WIDTH >30 km FROM SILDET ARE NOT CONSIDERED IN THE STATISTICS

Region and month	Lead category	Lead length		Lead area	
		($\times 10^3$km)	Contribution (%)	($\times 10^3$km^2)	Contribution (%)
Arctic Ocean in January	A	166.62	13.76	9.97	1.10
	B	613.54	50.66	166.87	18.43
	C	226.60	18.71	159.68	17.63
	D	181.41	14.98	361.50	39.92
	E	23.04	1.90	207.61	22.92
	Total	**1211.20**		**905.67**	
Arctic Ocean in April	A	230.92	13.41	13.81	1.06
	B	856.50	49.73	234.72	18.02
	C	334.03	19.39	235.36	18.07
	D	268.92	15.61	538.08	41.31
	E	32.07	1.86	280.48	21.53
	Total	**1722.45**		**1302.45**	
Beaufort Sea in January	A	31.53	12.14	1.89	0.77
	B	117.43	45.23	32.28	13.16
	C	51.01	19.65	36.36	14.83
	D	52.11	20.07	108.28	44.16
	E	7.55	2.91	66.39	27.08
	Total	**259.63**		**245.20**	
Beaufort Sea in April	A	39.66	14.22	2.36	1.05
	B	140.74	50.46	38.44	17.11
	C	51.72	18.54	36.30	16.16
	D	39.34	14.11	79.80	35.52
	E	7.62	2.73	68.62	30.54
	Total	**278.90**		**224.67**	

respectively. The contributions to total lead length and area for the different lead categories are similar for the two months. The highest contribution to the total length is from Category B, with about 50%. The contribution of Category A length to the total length in January and April is about 13%. In addition, wide leads (Category E) account for only 2% of the total lead length. As for the area, the contribution of Category D to the total area is the highest among all the lead categories, at about 40%. Category A leads account for only 1%, which is the smallest value in Table VI.

In the Beaufort Sea, a higher total lead length but a lower total lead area is found in April, compared with January. However, the contributions of the lead categories to lead length and area are small between January and April. Category D makes the most contribution to the total lead area, with more than 35%. The second highest and lowest contributions belong to categories E and A, respectively, which is consistent with the Arctic.

V. DISCUSSION

A. Ablation Experiments for SILDET

To evaluate the effectiveness of each module within SILDET, we conducted ablation experiments with various settings, including the loss function (CE loss and the loss function in the balance module) and the selection of network (i.e., PSPNet, U-Net, and DeepLabv3+) in the segmentation module, with and without the optimization module. Given that the mask module is a necessary component in the segmentation task based on remote sensing data, we used this module in all the experiments. All the experiments were evaluated with the scenes in the evaluation dataset, and the results are presented in Table VII. SILDET performs the best in all the experiments across all the metrics, indicating the importance of each module. In the following, we compare the different experiments to evaluate the role of the settings for each module in SILDET.

First, to evaluate the balance module, we compare Experiments 1 and 4. The segmentation network (PSPNet) was the same in both cases, but it was trained by the widely applied CE loss in Experiment 1 and by the loss function in the balance module in Experiment 4. By training the model with the loss function in the balance module, the detection of the open lead class, which accounts for a relatively small proportion of the training dataset, is improved. In Experiment 4, more open leads are detected and the recall is increased by 4%.

Second, we compared three deep learning networks in the segmentation module, i.e., U-Net [27], DeepLabv3+ [65], and PSPNet, which were used in Experiments 2–4, respectively. DeepLabv3+ is referred to as "DeepLab" for simplicity in Table VII. All the networks were trained by the loss function in the balance module. In addition, the optimization

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TABLE VII
RESULTS OF THE ABLATION STUDY ON THE OVERALL ARCHITECTURE OF SILDET

No.	Experiment	Precision (%)		Recall (%)		Kappa coefficient	Overall accuracy (%)
		Open lead	Frozen lead	Open lead	Frozen lead		
1	PSPNet+CEL+MM	84.14	91.93	86.56	77.43	0.85	97.45
2	U-Net+BM+MM	86.63	75.43	86.80	77.35	0.81	96.50
3	DeepLab+BM+MM	84.35	70.97	84.30	77.32	0.79	96.05
4	PSPNet+BM+MM	83.17	89.67	90.58	78.72	0.86	97.47
5	U-Net+BM+OM+MM	86.63	88.58	86.80	76.62	0.85	97.31
6	DeepLab+BM+OM+MM	84.35	81.95	84.30	78.39	0.82	93.75
7	PSPNet+BM+OM+MM (SILDET)	83.17	95.66	90.58	79.05	0.88	97.80

BM: balance module; OM: optimization module; MM: mask module.

TABLE VIII
LIST OF THE SCENES USED FOR ANALYZING THE DEPENDENCE OF THE INCIDENCE ANGLE

No.	Sentinel-1 product folder name
1	S1A_EW_GRDM_1SDH_20160107T171413_20160107T171513_009391_00D988_BDB5
2	S1A_EW_GRDM_1SDH_20160201T161634_20160201T161734_009755_00E419_7921
3	S1A_EW_GRDM_1SDH_20160308T161634_20160308T161728_010280_00F33D_39DF
4	S1A_EW_GRDM_1SDH_20160401T161530_20160401T161634_010630_00FD37_28C4
5	S1B_EW_GRDM_1SDH_20170102T152714_20170102T152818_003671_0064CA_64F8
6	S1B_EW_GRDM_1SDH_20170203T155932_20170203T160032_004138_007290_D988
7	S1A_EW_GRDM_1SDH_20170307T154444_20170307T154544_015588_019A0A_D03D
8	S1A_EW_GRDM_1SDH_20170401T162433_20170401T162533_015953_01A4E5_02C4
9	S1A_EW_GRDM_1SDH_20180101T154334_20180101T154434_019963_021FF4_86B4
10	S1A_EW_GRDM_1SDH_20180201T153414_20180201T153518_020415_022E3A_4B90
11	S1A_EW_GRDM_1SDH_20180307T155244_20180307T155323_020911_023E08_BC84
12	S1A_EW_GRDM_1SDH_20180401T163354_20180401T163454_021276_024997_8F71
13	S1B_EW_GRDM_1SDH_20190102T154244_20190102T154356_014317_01AA34_6915
14	S1B_EW_GRDM_1SDH_20190202T153535_20190202T153635_014769_01B8BB_5737
15	S1B_EW_GRDM_1SDH_20190304T162437_20190304T162537_015207_01C72D_3F0A
16	S1B_EW_GRDM_1SDH_20190403T153535_20190403T153635_015644_01D566_42FB
17	S1A_EW_GRDM_1SDH_20200103T154243_20200103T154347_030638_0382CE_5CB0
18	S1A_EW_GRDM_1SDH_20200201T155257_20200201T155357_031061_039187_CE33
19	S1A_EW_GRDM_1SDH_20200303T154446_20200303T154549_031513_03A144_77FC
20	S1B_EW_GRDM_1SDH_20200405T160824_20200405T160936_021011_027DB2_4B73
21	S1B_EW_GRDM_1SDH_20210102T164114_20210102T164214_024978_02F911_A284
22	S1A_EW_GRDM_1SDH_20210202T154452_20210202T154556_036413_044633_CB32
23	S1B_EW_GRDM_1SDH_20210303T164012_20210303T164112_025853_03155D_1D93
24	S1A_EW_GRDM_1SDH_20210403T154452_20210403T154556_037288_046499_CB4E

module was not used in Experiments 2–4. Experiment 4 using PSPNet achieves a Kappa coefficient improvement of 0.05–0.07, compared to the experiments based on the other networks. PSPNet shows the highest precision (90.58%) and recall (78.72%) for the frozen lead class. The detection performance using U-Net is slightly better than that for DeepLabv3+.

Finally, the contribution of the optimization module in SILDET could be estimated by comparing Experiments 4 and 7. Both experiments used PSPNet in the segmentation module and balance module. The optimization module was added in Experiment 7. This module improves the precision of frozen leads by filtering some incorrect regions. The results show that the optimization module significantly improves the precision for the frozen lead class by 6% and the Kappa coefficient by 0.02 in SILDET. Experiments 2 and 3 used U-Net and DeepLabv3+, respectively, in the segmentation module as well as the balance module. It was the same for Experiments 5–6 (respectively), but the optimization module was used in these two cases. The comparison of Experiments 2–3 and 5–6 demonstrates the effectiveness of the optimization module when other deep learning networks are used, with an increasing precision and Kappa coefficient for the frozen leads. These results indicate that the optimization module of SILDET utilizing the linear feature of leads can contribute to improved lead detection.

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TABLE IX
LIST OF THE SCENES IN THE TRAINING DATASET

No.	Sentinel-1 product folder name	Number of samples	Location
1	S1A_EW_GRDM_1SDH_20160103T174708_20160103T174802_009333_00D7DD_A700	10	Chukchi Sea
2	S1A_EW_GRDM_1SDH_20160103T192711_20160103T192811_009334_00D7E7_67EE	5	East Siberian Sea
3	S1A_EW_GRDM_1SDH_20160107T171413_20160107T171513_009391_00D988_BDB5	13	Beaufort Sea
4	S1A_EW_GRDM_1SDH_20160111T164118_20160111T164212_009449_00DB2D_1988	9	Beaufort Sea
5	S1A_EW_GRDM_1SDH_20160201T161634_20160201T161734_009755_00E419_7921	15	Beaufort Sea
6	S1A_EW_GRDM_1SDH_20160203T160008_20160203T160102_009784_00E4EE_A332	10	Beaufort Sea
7	S1A_EW_GRDM_1SDH_20160204T164117_20160204T164211_009799_00E55C_B7A7	9	Beaufort Sea
8	S1A_EW_GRDM_1SDH_20160206T080903_20160206T081003_009823_00E60C_FF30	5	Greenland Sea
9	S1A_EW_GRDM_1SDH_20160207T021616_20160207T021720_009834_00E65B_8A29	6	Kara Sea
10	S1A_EW_GRDM_1SDH_20160306T195047_20160306T195147_010253_00F276_BE66	10	East Siberian Sea
11	S1A_EW_GRDM_1SDH_20160308T161529_20160308T161634_010280_00F33D_0575	10	Beaufort Sea
12	S1A_EW_GRDM_1SDH_20160308T161634_20160308T161728_010280_00F33D_39DF	12	Beaufort Sea
13	S1A_EW_GRDM_1SDH_20160322T155904_20160322T160008_010484_00F902_8C96	10	Beaufort Sea
14	S1A_EW_GRDM_1SDH_20160401T161530_20160401T161634_010630_00FD37_28C4	14	Beaufort Sea
15	S1A_EW_GRDM_1SDH_20160404T050652_20160404T050757_010667_00FE52_1022	9	Central Arctic
16	S1A_EW_GRDM_1SDH_20160404T131748_20160404T131848_010672_00FE77_B689	5	Barents Sea
17	S1A_EW_GRDM_1SDH_20160409T151145_20160409T151245_010746_0100AB_B7FD	6	Maclure Strait
18	S1A_EW_GRDM_1SDH_20160415T155905_20160415T160009_010834_010353_88B6	11	Beaufort Sea
19	S1A_EW_GRDM_1SDH_20170411T150318_20170411T150418_016098_01A94F_9C77	4	Beaufort Sea
20	S1A_EW_GRDM_1SDH_20170416T151133_20170416T151233_016171_01AB86_25BD	10	Beaufort Sea
21	S1A_EW_GRDM_1SDH_20170424T154428_20170424T154528_016288_01AF20_587E	9	Beaufort Sea
22	S1A_EW_GRDM_1SSH_20160106T081351_20160106T081451_009371_00D8F4_44A9	10	Laptev Sea
23	S1A_EW_GRDM_1SSH_20160303T110532_20160303T110637_010204_00F107_4B24	9	Kara Sea

B. Method Constraints and Potential

SILDET is a lead detection algorithm which has applicability for SAR images containing open and frozen leads. However, there are four aspects of SILDET that should be further improved.

First, it is difficult to detect narrow leads with SILDET, especially leads with a width <360 m (see Section IV-B). One of the possible reasons for this could be the mixed pixels of narrow leads [15], [63]. When the lead area is equivalent to the area of one pixel, the lead is not always covered by a single pixel. An observation from a pixel covering a lead sometimes includes a contribution from surface types other than lead. This complicates the interpretation of the observation from a mixed pixel. Furthermore, SAR observations can contain noise, as well as overlap of the backscatter from the lead and other surface types. Methods based on a threshold for the backscatter cannot handle these situations. Compared with the threshold approach, deep learning networks incorporate more image information to improve the performance of detecting leads. Texture plays an important role in detecting leads in SAR images. However, a narrow lead usually covers a limited number of pixels in the image, allowing for less texture information. In addition, PSPNet in SILDET utilizes the top-most feature maps computed by ResNet-101, which shows a better performance than the other ResNets (Table X in Appendix F). The top feature maps have a large receptive field and rich semantic information but with a coarse resolution. There is little information left in the top-most features for small objects, which may compromise the segmentation performance for narrow leads. Improving the detection of small leads could allow more accurate lead distribution monitoring.

The second issue is the wind effect on the water surface in the leads, which can cause misclassification as ice [20], [28]. In this study, we did not consider this effect on open leads. However, the effect needs to be considered when the leads are very wide (a few kilometers). This issue could be resolved by

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TABLE X
RESULTS OF THE ABLATION STUDY ON THE BACKBONE FOR THE PSPNET OF SILDET

No.	Experiment	Precision (%)		Recall (%)		Kappa coefficient	Overall accuracy (%)
		Open lead	Frozen lead	Open lead	Frozen lead		
1	ResNet-34	84.85	91.67	86.68	77.73	0.86	97.46
2	ResNet-50	86.61	93.89	85.85	71.78	0.84	97.28
3	ResNet-101 (SILDET)	83.17	95.66	90.58	79.05	0.88	97.80

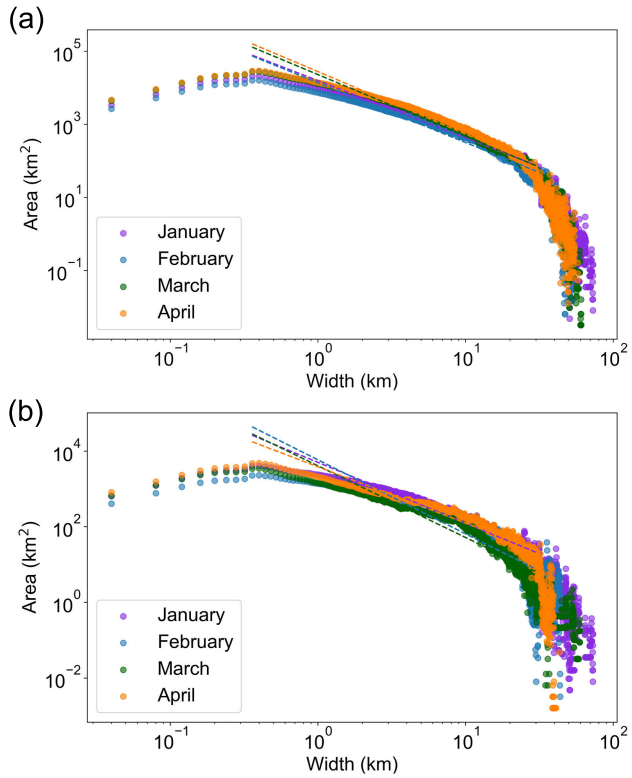


Fig. 10. Log–log plot of the lead width distribution from the SILDET results from January to April 2023 across (a) Arctic and (b) Beaufort Sea. The straight lines indicate the fit power-law curves. Note: the power-law fittings apply only for a lead width of 360 m–30 km.

utilizing cross-polarization SAR data [66]. Therefore, a strong and robust method of removing the thermal noise in cross-polarization images needs to be explored in the future.

Third, SILDET can misclassify some classes when their backscatter and texture are similar. For example, in Scene 8, the ice has a lower backscatter than the surrounding ice, resulting in its misclassification as an open lead. It is possible that the ice surface (whether bare or snow-covered) becomes wet. Furthermore, when a lead freezes while the ice develops under calm water conditions, it returns very low backscatter and smooth texture and hence may be misclassified as open lead, even though the ice becomes thick. Some physical conditions hinder accurate lead identification. A more comprehensive and balanced training dataset, with scenes covered by leads with different backscatter and texture, as well as different ice conditions, should allow a more comprehensive assessment of the performance of the network. Meanwhile, it is difficult to distinguish ridges and leads with frost flowers in SAR images. This can lead to an overestimation of Kappa and accuracy when ridges are classified as leads in the visual interpretation and SILDET results. Due to the lack of field data, we cannot consider these factors quantitatively.

Finally, in the preprocessing of Sentinel-1 images, we use a threshold to separate water and ice and determine the normalization coefficients for each type in the incidence angle normalization. Although this approach is simple and rapid, it may bring prior error information for SILDET. Meanwhile, SILDET may encounter difficulties when the same type of sea ice is located on different edges of the range swath of SAR images because the incidence angle dependence might not be normalized perfectly.

Based on SILDET, we derived the lead frequency distribution in the Arctic from January to April 2023. The regions with a relatively high lead frequency are consistent with previous studies [19], [67]. Meanwhile, the Arctic-wide lead width distribution follows a power law, with an average exponent of 1.65 from January to April 2023. This is in agreement with the results from previous studies, e.g., Wadhams [68] and Lindsay and Rothrock [69], which were, respectively, based on data from a submarine mission and Advanced Very High Resolution Radiometer (AVHRR) satellite data, with a power-law exponent of 2.0 and around 1.6, respectively. There was also a previous study based on the CryoSat-2 satellite which yielded an exponent of about 2.47 [15]. The difference in the exponents can be attributed to the different measurement systems (satellite and submarine), lead definitions, and detection methods.

The results presented in this article demonstrate the potential of SILDET for monitoring Arctic-wide sea ice leads during wintertime. However, as Sentinel-1B has been retired, the frequency of the Sentinel-1 coverage has decreased. In the future, the launch of the Sentinel-1C and Sentinel-1D satellites should enable more frequent coverage. In addition, the development of a cross-platform method for lead detection based on SAR images is necessary. This would provide extensive spatial and temporal coverage for Arctic sea ice lead monitoring based on the data from multiple satellites. Cross-calibration between SAR systems will be needed in this case.

VI. CONCLUSION

In this article, we have proposed an automatic method to detect open and newly frozen sea ice leads covered by thin ice in wintertime (from January to April) in the Arctic, using Sentinel-1 HH polarization images. The method, called SILDET, is composed of four modules: 1) a segmentation

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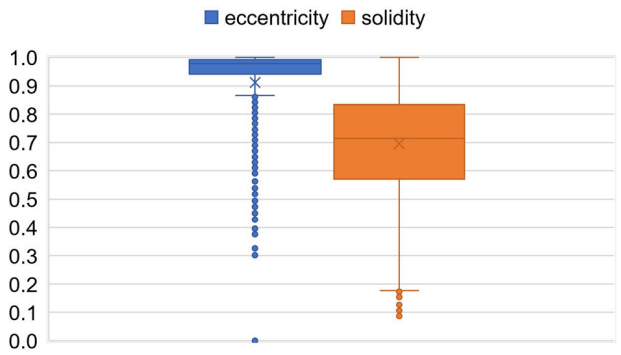


Fig. 11. Eccentricity of the frozen leads from the training dataset.

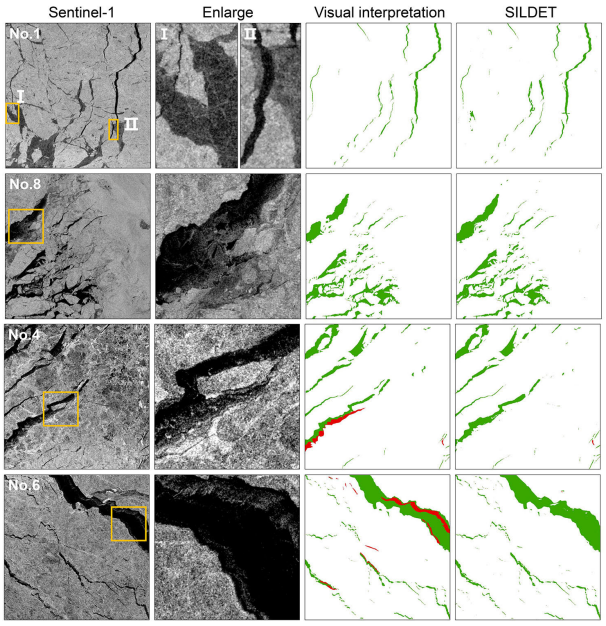


Fig. 12. Four Sentinel-1 σ^o_{hh} images of scenes (first column). Enlargements of the areas in the yellow boxes are shown in the second column. Lead identification based on visual interpretation of SAR images and the SILDET method is shown in the third and fourth columns, respectively. The color bar is consistent with that in Fig. 5.

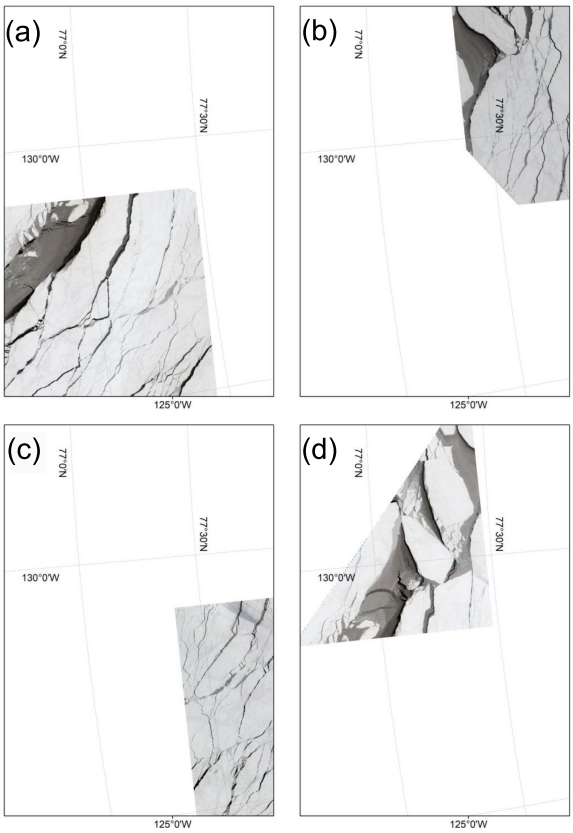


Fig. 13. Sentinel-2 scenes used in this study. The frame is the same as that in Fig. 6. The scenes were acquired at (a) 21:22 on April 20, 2019, (b) 22:31 on April 21, 2019, (c) 21:40 on April 21, 2019, and (d) 21:40 on April 21, 2019. The overlay order (from top to bottom) of the Sentinel-2 scenes in Fig. 6(b) is (a)–(d).

module; 2) a balance module; 3) an optimization module; and 4) a mask module. In the segmentation module, PSPNet utilizes global content to improve the accuracy of the detection and the generalization ability. A loss function is then introduced to alleviate the class imbalance problem in the balance module. In the optimization module, the shape features of leads are utilized to improve the precision evaluation for frozen leads. Land, landfast ice, and areas outside the ice cover are removed in the mask module.

By comparing the SILDET results with the lead results from visual interpretation of SAR imagery covering different ice conditions and development stages of leads, the OA and Kappa were 97.80% and 0.88, respectively. This shows that the sea ice lead maps detected by SILDET agree with the visual interpretation well. Furthermore, SILDET showed a good identification performance for both water and thin ice in the sea ice leads. SILDET achieved detection accuracy for the open lead class with an overall precision of 83.17% and recall of 90.58%. For the frozen lead class, the overall precision and recall from SILDET were 95.66% and 79.05%, respectively. The ablation experiments demonstrated that the segmentation module, balance module, and optimization module in SILDET contribute equally to the improvement of the detection performance. The SILDET detection results were compared with the lead maps from Sentinel-2 images, lead maps from an existing SAR-based lead detection method, and a lead dataset based on MODIS images. SILDET captured most of the open leads and frozen leads shown in the Sentinel-2 maps. Compared with the lead maps from the other two methods, the lead width distribution and area from SILDET were closer to those from Sentinel-2. We derived the lead frequency distribution in the Arctic from January to April 2023 using SILDET and analyzed the lead width distribution. The Arctic lead width derived from the SILDET detection result followed a power-law distribution, with the fit exponents varying from 1.57 to 1.71. Overall, SILDET is an effective and robust method for the automated detection of Arctic sea ice leads with their open and frozen components in wintertime using Sentinel-1 HH polarization imagery. The method can be used as a tool to monitor sea ice lead distribution at a fine resolution.

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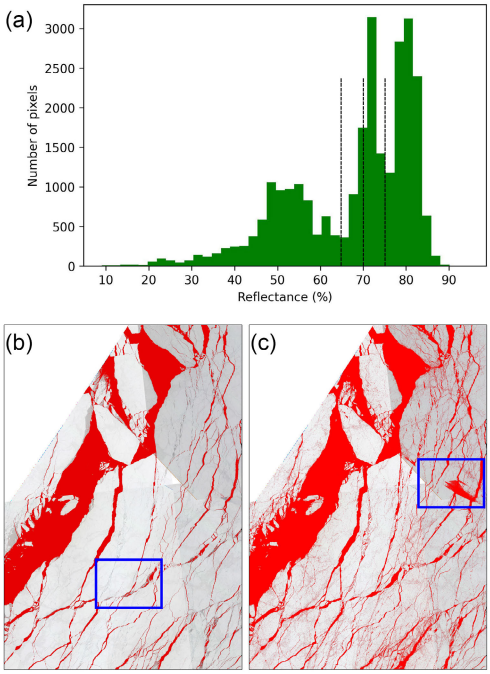


Fig. 14. (a) Pixel histogram of the reflectance of the Sentinel-2 band 2 images; the dashed lines correspond to the three thresholds of 65, 70, and 75. (b) and (c) Binary images for threshold =65 and threshold =75, respectively.

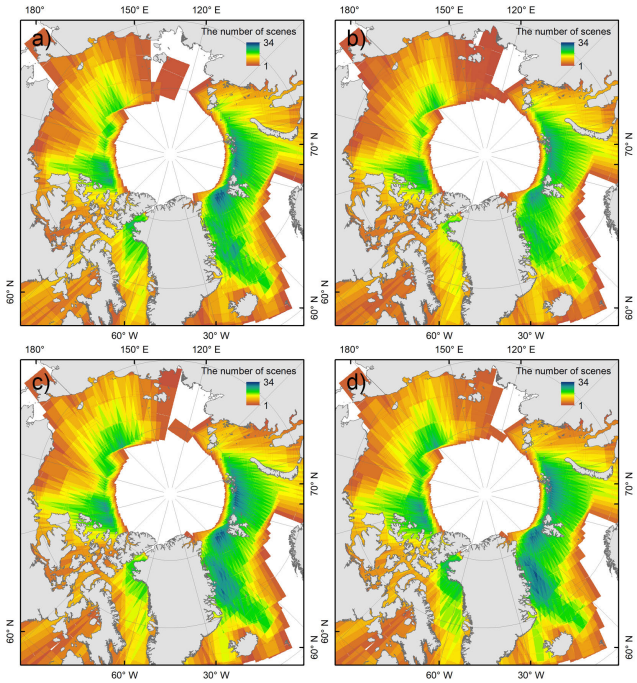


Fig. 15. (a)–(d) Number of Sentinel-1 scenes in the Arctic from January to April 2023. Note that the white areas in the figure indicate no Sentinel-1 data.

APPENDIX

A. Lists of the Scenes Used for Analyzing the Dependence of the Incidence Angle and the Training Dataset

The lists of the scenes used for analyzing the dependence of the incidence angle and the training dataset are provided in Tables VIII and IX, respectively.

B. Feature Thresholds for the Refinement Module

Fig. 11 shows the eccentricity and solidity distribution of the frozen leads from the training dataset. We found that the eccentricity and solidity of most of the objects are higher than 0.95 and 0.55, respectively. Therefore, the objects with eccentricity <0.95 and solidity >0.55 are not considered to be frozen leads.

C. SILDET Results and Visual Interpretation of the Scenes (Nos. 1, 8, 4, and 6)

Fig. 12 shows the SILDET results and visual interpretation of the scenes (Nos. 1, 8, 4, and 6).

D. Locations of the Sentinel-2 Images Used in the Study and the Threshold for Generating Lead Maps

The locations of each Sentinel-2 scene used in the study are shown in Fig. 13.

The selection of the threshold for generating lead maps from Sentinel-2 images is shown in Fig. 13. The lead maps corresponding to the thresholds of 65% and 75% are shown in Fig. 14(b) and (c) with incomplete leads and cloud contamination, respectively (marked by the blue boxes).

E. Number of Sentinel-1 Scenes Used for Fig. 8

This is shown in Fig. 15.

F. Ablation Experiment for the Backbone of PSPNet in SILDET

This is shown in Table X.

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REFERENCES

[1] W. M. Organization. *Sea-Ice Nomenclature*. Accessed: Jan. 18, 2024. [Online]. Available: https://library.wmo.int/index.php?lvl=notice_display&id=6772

[2] T. Fichefet and M. Maqueda, "MA: On modelling the sea-ice-ocean system," in *Problemes Non Lineaires Appliques*, vol. 1995. France: Le Havre, 1994, pp. 343–420.

[3] G. A. Maykut, "Energy exchange over young sea ice in the central Arctic," *J. Geophys. Res., Oceans*, vol. 83, no. C7, pp. 3646–3658, Jul. 1978.

[4] A. Alam and J. A. Curry, "Evolution of new ice and turbulent fluxes over freezing winter leads," *J. Geophys. Res., Oceans*, vol. 103, no. C8, pp. 15783–15802, Jul. 1998.

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[5] J. O. Pinto, A. Alam, J. Maslanik, J. Curry, and R. S. Stone, "Surface characteristics and atmospheric footprint of springtime Arctic leads at SHEBA," *J. Geophys. Res., Oceans*, vol. 108, no. C4, p. 8051, 2003.

[6] P. Rampal, J. Weiss, and D. Marsan, "Positive trend in the mean speed and deformation rate of Arctic sea ice, 1979–2007," *J. Geophys. Res., Oceans*, vol. 114, no. C5, 2009.

[7] G. Spreen, R. Kwok, and D. Menemenlis, "Trends in Arctic sea ice drift and role of wind forcing: 1992–2009," *Geophys. Res. Lett.*, vol. 38, no. 19, 2011, Art. no. L19501.

[8] B. J. Lewis and J. K. Hutchings, "Leads and associated sea ice drift in the Beaufort sea in winter," *J. Geophys. Res., Oceans*, vol. 124, no. 5, pp. 3411–3427, May 2019.

[9] M. Li, C. Q. Ke, X. Shen, B. Cheng, and H. Li, "Investigation of the Arctic Sea ice volume from 2002 to 2018 using multi-source data," *Int. J. Climatol.*, vol. 41, no. 4, pp. 2509–2527, 2021, doi: https://doi.org/0.1002/joc.6972

[10] Y. Zhang, X. Cheng, J. Liu, and F. Hui, "The potential of sea ice leads as a predictor for summer Arctic sea ice extent," *Cryosphere*, vol. 12, no. 12, pp. 3747–3757, 2018.

[11] F. Reiser, S. Willmes, and G. Heinemann, "A new algorithm for daily sea ice lead identification in the Arctic and Antarctic winter from thermal-infrared satellite imagery," *Remote Sens.*, vol. 12, no. 12, p. 1957, Jun. 2020.

[12] J. P. Hoffman, S. A. Ackerman, Y. Liu, J. R. Key, and I. L. McConnell, "Application of a convolutional neural network for the detection of sea ice leads," *Remote Sens.*, vol. 13, no. 22, p. 4571, Nov. 2021.

[13] J. Röhrs and L. Kaleschke, "An algorithm to detect sea ice leads by using AMSR-E passive microwave imagery," *Cryosphere*, vol. 6, no. 2, pp. 343–352, 2012.

[14] M. Li, J. Liu, M. Qu, Z. Zhang, and X. Liang, "An analysis of Arctic sea ice leads retrieved from AMSR-E/AMSR2," *Remote Sens.*, vol. 14, no. 4, p. 969, Feb. 2022.

[15] A. Wernecke and L. Kaleschke, "Lead detection in Arctic sea ice from CryoSat-2: Quality assessment, lead area fraction and width distribution," *Cryosphere*, vol. 9, no. 5, pp. 1955–1968, 2015.

[16] D. Bröhan and L. Kaleschke, "A nine-year climatology of Arctic sea ice lead orientation and frequency from AMSR-E," *Remote Sens.*, vol. 6, no. 2, pp. 1451–1475, 2014.

[17] N. Ivanova, P. Rampal, and S. Bouillon, "Error assessment of satellite-derived lead fraction in the Arctic," *Cryosphere*, vol. 10, no. 2, pp. 585–595, 2016.

[18] M. Qu, X. Pang, X. Zhao, J. Zhang, Q. Ji, and P. Fan, "Estimation of turbulent heat flux over leads using satellite thermal images," *Cryosphere*, vol. 13, no. 6, pp. 1565–1582, Jun. 2019.

[19] J. P. Hoffman, S. A. Ackerman, Y. Liu, and J. R. Key, "A 20-year climatology of sea ice leads detected in infrared satellite imagery using a convolutional neural network," *Remote Sens.*, vol. 14, no. 22, p. 5763, Nov. 2022.

[20] D. Murashkin, G. Spreen, M. Huntemann, and W. Dierking, "Method for detection of leads from Sentinel-1 SAR images," *Ann. Glaciol.*, vol. 59, pp. 124–136, Jul. 2018.

[21] D. Murashkin and G. Spreen, "Sea ice leads detected from Sentinel-1 SAR images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 174–177.

[22] L. Ma, Y. Liu, X. Zhang, Y. Ye, G. Yin, and B. A. Johnson, "Deep learning in remote sensing applications: A meta-analysis and review," *ISPRS J. Photogramm. Remote Sens.*, vol. 152, pp. 166–177, Jun. 2019.

[23] X. Li et al., "Deep-learning-based information mining from ocean remote-sensing imagery," *Nat. Sci. Rev.*, vol. 7, no. 10, pp. 1584–1605, Oct. 2020.

[24] D. Murashkin and A. Frost, "Arctic sea ICE mapping using Sentinel-1 SAR scenes with a convolutional neural network," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 5660–5663.

[25] N. Zakhvatkina, V. Smirnov, I. Bychkova, and V. Stepanov, "Detection of the leads in the Arctic drifting sea ice on SAR images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 4276–4279.

[26] D. Murashkin, "Remote sensing of sea ice leads with Sentinel-1 C-band synthetic aperture radar," Ph.D. dissertation, Inst. Environ. Phys., Univ. Bremen, 2024.

[27] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent. (MICCAI)*. Munich, Germany: Springer, Oct. 2015, pp. 234–241.

[28] Z. Liang, X. Pang, Q. Ji, X. Zhao, G. Li, and Y. Chen, "An entropy-weighted network for polar sea ice open lead detection from Sentinel-1 SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4304714.

[29] S. Wang and X. Yao, "Multiclass imbalance problems: Analysis and potential solutions," *IEEE Trans. Syst., Man, Cybern., B (Cybern.)*, vol. 42, no. 4, pp. 1119–1130, Aug. 2012.

[30] M. Buda, A. Maki, and M. A. Mazurowski, "A systematic study of the class imbalance problem in convolutional neural networks," *Neural Netw.*, vol. 106, pp. 249–259, Oct. 2018.

[31] S. V. Nghiem et al., "Evolution in polarimetric signatures of thin saline ice under constant growth," *Radio Sci.*, vol. 32, no. 1, pp. 127–151, Jan. 1997.

[32] D. Isleifson, B. Hwang, D. G. Barber, R. K. Scharien, and L. Shafai, "C-band polarimetric backscattering signatures of newly formed sea ice during fall freeze-up," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 8, pp. 3256–3267, Aug. 2010.

[33] M. Shokr and M. Dabboor, "Observations of SAR polarimetric parameters of lake and fast sea ice during the early growth phase," *Remote Sens. Environ.*, vol. 247, Sep. 2020, Art. no. 111910.

[34] M. Shokr and N. K. Sinha, *Sea Ice: Physics and Remote Sensing*. Hoboken, NJ, USA: Wiley, 2023.

[35] D. Isleifson, R. J. Galley, D. G. Barber, J. C. Landy, A. S. Komarov, and L. Shafai, "A study on the C-band polarimetric scattering and physical characteristics of frost flowers on experimental sea ice," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 3, pp. 1787–1798, Mar. 2014.

[36] J. Lohse, A. P. Doulgeris, and W. Dierking, "Incident angle dependence of Sentinel-1 texture features for sea ice classification," *Remote Sens.*, vol. 13, no. 4, p. 552, Feb. 2021.

[37] J.-W. Park, A. A. Korosov, M. Babiker, S. Sandven, and J.-S. Won, "Efficient thermal noise removal for Sentinel-1 TOPSAR cross-polarization channel," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1555–1565, Mar. 2018.

[38] Y. Sun and X.-M. Li, "Denoising Sentinel-1 extra-wide mode cross-polarization images over sea ice," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2116–2131, Mar. 2021.

[39] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.

[40] G. Spreen, L. Kaleschke, and G. Heygster, "Sea ice remote sensing using AMSR-E 89-GHz channels," *J. Geophys. Res., Oceans*, vol. 113, no. C2, 2008, Art. no. C02S03, doi: 10.1029/2005JC003384.

[41] H. Hersbach et al., "ERA5 hourly data on single levels from 1979 to present," *Copernicus Climate Change Service (C3S) Climate Data Store (CDS)*, vol. 10, no. 10, 2018, Art. no. 024381.

[42] L. von Albedyll et al., "Thermodynamic and dynamic contributions to seasonal Arctic sea ice thickness distributions from airborne observations," *Elementa: Sci. Anthropocene*, vol. 10, no. 1, p. 00074, 2022.

[43] L. von Albedyll et al., "Lead fractions from SAR-derived sea ice divergence during MOSAiC," *Cryosphere Discuss.*, vol. 2023, pp. 1–39, Aug. 2023.

[44] M. Qu et al., "Spring leads in the Beaufort sea and its interannual trend using Terra/MODIS thermal imagery," *Remote Sens. Environ.*, vol. 256, Apr. 2021, Art. no. 112342.

[45] M. S. Mahmud, T. Geldsetzer, S. E. L. Howell, J. J. Yackel, V. Nandan, and R. K. Scharien, "Incidence angle dependence of HH-polarized C- and L-band wintertime backscatter over Arctic sea ice," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6686–6698, Nov. 2018.

[46] W. Guo, P. Itkin, J. Lohse, M. Johansson, and A. P. Doulgeris, "Cross-platform classification of level and deformed sea ice considering per-class incident angle dependency of backscatter intensity," *Cryosphere*, vol. 16, no. 1, pp. 237–257, 2022.

[47] A. S. Komarov, V. Zabeline, and D. G. Barber, "Ocean surface wind speed retrieval from C-band SAR images without wind direction input," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 2, pp. 980–990, Feb. 2014.

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[48] W. M. Organization, *WMO Sea-Ice Nomenclature: Terminology, Codes and Illustrated Glossary*, Secretariat of the World meteorological organization, Geneva, Switzerland, 1970.
[49] S. Beaven and S. Gogineni, "Shipborne radar backscatter measurements from Arctic sea ice during the fall freeze-up," *Remote Sens. Rev.*, vol. 9, nos. 1–2, pp. 3–25, 1994.
[50] S. Beaven, S. Gogineni, and M. Shanableh, "Radar backscatter signatures of thin sea ice in the central Arctic," *Int. J. Remote Sens.*, vol. 15, no. 5, pp. 1149–1154, 1994.
[51] D. Kinga and J. B. Adam, "A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, vol. 5, San Diego, CA, USA, 2015, p. 6.
[52] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[53] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. 4th Int. Conf. 3D Vis. (3DV)*, Oct. 2016, pp. 565–571.
[54] A. Shrivastava, A. Gupta, and R. Girshick, "Training region-based object detectors with online hard example mining," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 761–769.
[55] F. P. Preparata, "The medial axis of a simple polygon," in *Proc. 6th Symp. Math. Found. Comput. Sci.* Slovakia: Springer, Sep. 1977, pp. 443–450.
[56] N. Zakhvatkina, A. Korosov, S. Muckenhuber, S. Sandven, and M. Babiker, "Operational algorithm for ice–water classification on dual-polarized RADARSAT-2 images," *Cryosphere*, vol. 11, no. 1, pp. 33–46, 2017.
[57] M. Story and R. G. Congalton, "Accuracy assessment: A user's perspective," *Photogramm. Eng. Remote Sens.*, vol. 52, no. 3, pp. 397–399, 1986.
[58] R. G. Congalton, "A review of assessing the accuracy of classifications of remotely sensed data," *Remote Sens. Environ.*, vol. 37, no. 1, pp. 35–46, Jul. 1991.
[59] J. Cohen, "A coefficient of agreement for nominal scales," *Educ. Psychol. Meas.*, vol. 20, no. 1, pp. 37–46, Apr. 1960.
[60] P. Wadhams, "The underside of Arctic sea ice imaged by sidescan sonar," *Nature*, vol. 333, no. 6169, pp. 161–164, May 1988.
[61] D. Marsan, H. Stern, R. Lindsay, and J. Weiss, "Scale dependence and localization of the deformation of Arctic sea ice," *Phys. Rev. Lett.*, vol. 93, no. 17, Oct. 2004, Art. no. 178501.
[62] H. Stern and R. Lindsay, "Spatial scaling of Arctic sea ice deformation," *J. Geophys. Res., Oceans*, vol. 114, no. C10, 2009, Art. no. C10017, doi: 10.1029/2009JC005380.
[63] M. Muchow, A. U. Schmitt, and L. Kaleschke, "A lead-width distribution for Antarctic sea ice: A case study for the Weddell Sea with high-resolution Sentinel-2 images," *Cryosphere*, vol. 15, no. 9, pp. 4527–4537, 2021.
[64] L. Thielke, G. Spreen, M. Huntemann, and D. Murashkin, "Spatio-temporal variability of small-scale leads based on helicopter maps of winter sea ice surface temperatures," *Elementa: Sci. Anthropocene*, vol. 12, no. 1, p. 00023, 2024, doi: 10.1525/elementa.2023.00023.
[65] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.
[66] Y. Lu, B. Zhang, and W. Perrie, "Arctic sea ice and open water classification from spaceborne fully polarimetric synthetic aperture radar," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4203713.
[67] S. Willmes, G. Heinemann, and F. Schnaase, "Patterns of wintertime Arctic sea-ice leads and their relation to winds and ocean currents," *Cryosphere*, vol. 17, no. 8, pp. 3291–3308, Aug. 2023.
[68] P. Wadhams, "Sea-ice topography of the Arctic Ocean in the region 70 W to 25 E," *Philos. Trans. Roy. Soc. London. A, Math. Phys. Sci.*, vol. 302, no. 1464, pp. 45–85, 1981.
[69] R. Lindsay and D. Rothrock, "Arctic sea ice leads from advanced very high resolution radiometer images," *J. Geophys. Res., Oceans*, vol. 100, no. C3, pp. 4533–4544, 1995.



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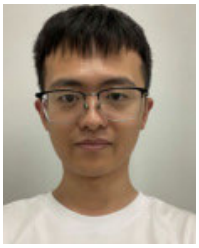
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